

ARTICLE



Developing stealth assessments to assess young Chinese learners' L2 reading comprehension

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Abstract

This study explored the use of stealth assessments within a digital game to assess second language (L2) Chinese learners' reading comprehension. Log data tracking learners' in-game behaviors from a game designed for Chinese dual language immersion classrooms (Poole et al., 2022) were used to construct Bayesian Belief Networks to model reading comprehension. Variables included the use of an in-game glossary for word lookups, time spent reading texts, vocabulary knowledge, text length, response requirements, and the number of repetitions. Networks were first built using class-level gameplay data and subsequently applied to individual students. Results showed significant correlations between students' modeled comprehension and their performance on an external reading comprehension assessment ($r = .52$) as well as teachers' informal reading evaluations ($r = .66$), demonstrating stealth assessments' potential for unobtrusive reading comprehension measurement. The findings contribute to the understanding of digital game-based learning and assessment in language education, particularly in the context of reading comprehension for non-alphabetic languages like Mandarin Chinese.

Keywords: *Stealth Assessment; Bayesian Belief Networks; Reading Comprehension; Dual Language Immersion; Less commonly taught languages*

Language(s) Learned in This Study: *Mandarin Chinese*

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Introduction

One of the often overlooked and under-researched advantages of utilizing digital games for additional language (L2) learning is their potential to assess learners while they engage with the target language in highly contextualized environments, without the explicit pretext of an assessment (Mavridis & Tsiatsos, 2017). Many assessments (for example, fill-in-the-blank, multiple-choice items) have been criticized for their lack of contextualization (Shute & Wang, 2015) and isolated evaluation processes through the pre-post assessment paradigm (Clarke-Midura & Dede, 2010). In contrast, digital games can be leveraged to track learner actions, in-game choices, and behavioral patterns which may act as indicators of L2 proficiency and/or development. Perhaps more importantly, data collected during gameplay provide multiple observations of student performance, which is essential for creating robust assessments of learning and competency (Clarke-Midura & Dede, 2010). Furthermore, game-based measures can be collected *while*, not *after*, learners engage in meaningful, contextualized tasks. In other educational areas (for example, mathematics and biology), researchers have utilized log data from digital games to collect evidence linking in-game behavioral actions with cognitive choices (Gibson & Clarke-Midura, 2015), changes in conceptual understanding (Martin et al., 2015), and affective responses (Halverson & Owen,

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2014). However, this approach has rarely been applied to studies assessing L2 proficiency or performance.

In the present study, game-based assessment approaches inspired by those used in other educational settings were applied to an L2 learning context. Using a digital game designed to promote vocabulary learning and reading development for students in a Chinese dual language immersion (DLI) classroom (Poole et al., 2022), a variety of student data were gathered, including frequency of vocabulary support use, reading in-game texts, and gameplay behaviors. Data were used to construct Bayesian Belief Networks (BBNs) to assess learners' reading comprehension. Our findings provide evidence that game-based assessments can reliably measure language learners' reading comprehension in L2 Mandarin Chinese (hereafter referred to as Chinese).

Literature Review

Reading in (L2) Chinese

Reading is the process of extracting information from written text which involves rapid decoding and integration of orthographic, phonological, semantic, morphological, and cultural knowledge. Both monolingual and multilingual readers rely on prior knowledge of the world and language(s) to engage in bottom-up (that is, combining graphemes and morphemes into words, clauses, and sentences) and top-down (that is, relying on prior knowledge to support comprehension) processing when reading (Koda, 2007). Bernhardt (2011) emphasized that L2 readers' comprehension ability comprises their L1 literacy skills, L2 knowledge, and variance which may include background knowledge, life experiences, and metacognitive strategies. Importantly, L2 readers are different from L1 readers in that their repertoires of oral language and overall vocabulary sizes are usually smaller (especially at novice and intermediate levels of proficiency, Grabe, 2014), typically resulting in more bottom-up processing (Koda, 2007). Additionally, reading often involves both lower-level processing (that is, more superficial word, phrase, and sentence processing) and higher-level processing (that is, textual meaning integration, inferencing, predictions, and so forth). Novice L2 readers spend much of their time using bottom-up strategies for lower-level processing (Grabe, 2014). While researchers generally agree that L2 readers will engage in a mixture of processes (Alderson et al., 2015), given that novice learners are generally focused on decoding words, their high-level processing is typically limited.

Chinese presents an interesting (and challenging) case for L2 readers as a morphographic language in which written characters correspond to morphemes, with most words composed of multiple, typically two, characters (Li, 2019). For example, 电 (diàn, 'electricity') appears in words like 电脑 ('electric-brain' meaning computer) and 电梯 ('electric-ladder', meaning elevator). Chinese characters are typically made up of multiple components that provide information about a character's pronunciation ('phonetic components') and/or meaning ('semantic components'). However, these components' functions are not always systematic or transparent (Everson, 2016). While L1 readers, given their large repertoire of characters from which to extrapolate patterns and make connections, are often able to make use of these components while reading unknown characters and words, they vary in their ability to do this (Everson, 2016). The unreliability of character components compounds the challenge of reading for L2 readers of Chinese who are not readily able to identify links between print and oral language and are therefore less able to use their oral language repertoires when reading (Dew, 1994). However, given that Chinese characters are made up of a finite number of recurring components and that the majority of these components do relate functionally to the characters in which they appear (Coss, 2023) when assessing L2 readers' reading abilities and vocabulary development, it is important to document the development of partial word knowledge accumulation, including understanding when, how, and why learners rely on sub-character (that is, component) cues while reading. This is because proficient use of character components is a feature of strong reading comprehension ability in L2 learners (Shen & Ke, 2007). However, especially for emerging L2 readers, Chinese characters and texts are often glossed with more readily

decodable representations of their pronunciation, most typically Hanyu Pinyin (hereafter referred to as pinyin).

Pinyin uses the Roman alphabet with four diacritic markers to represent syllabic (including tonal) pronunciation information for Mandarin Chinese. It was created for L1 Chinese readers to foster literacy development as a systematic linking of characters to oral language (Li, 2019). Pinyin is often used as a scaffold for reading for L2 readers as well. Further, Lü (2017) highlighted the role of pinyin as a tool for bridging oral and written Chinese, especially for beginners who struggle with character recognition. By providing phonetic information, pinyin allows learners to leverage their oral proficiency to facilitate reading comprehension, and is a useful resource, especially in the early stages of L1 literacy development as well as L2 learning.

Research has highlighted three skills that proficient L2 readers of Chinese possess: automatized character recognition, word segmentation ability, and character component awareness (Shen & Jiang, 2013). These skills are relevant to the present study for three reasons. First, when designing a digital game for L2 Chinese learners, one must consider both the oral proficiency and character recognition/reading abilities of the learners. Unlike learners of additional languages with alphabetic scripts, Chinese L2 learners' ability to rely on their oral language for textual decoding while reading is limited, especially at lower levels of proficiency (Everson, 2011). The game used in the present study primarily relies on characters and words that students already know orally, with the goal of reducing cognitive load during reading. Secondly, it is important to provide learners with support to address the challenges of reading Chinese characters in context. In the present study, in-text glosses are provided for characters identified as potentially difficult for the learners. Glosses provide on-demand information about a word (or character) without disrupting the reading process. Past research has shown that glossing can help improve reading fluency, enhance reading comprehension (Hong, 1997), and reduce reading anxiety (Zhao et al., 2013). This study used a glossing system that allowed learners to access pinyin for any Chinese character encountered throughout the game (see Figure 1).

Figure 1

Glossing Tool



(Stealth) Assessments in Games and Virtual Environments

Although research on the design and evaluation of assessments in games and virtual environments is still emerging, studies have explored various in-game variables to predict learning, such as time, frequency, sequence of choice, and user path selections (Halverson & Owen, 2014; Liu et al., 2016; Shute & Wang, 2015). Some studies have explored how log data, like time on task, in-game sequences, and playstyles, relate to language learning outcomes (Collentine, 2011; Hsiao et al., 2017; Hwang et al., 2017). However, there is a lack of research exploring how to leverage this data for L2 assessments.

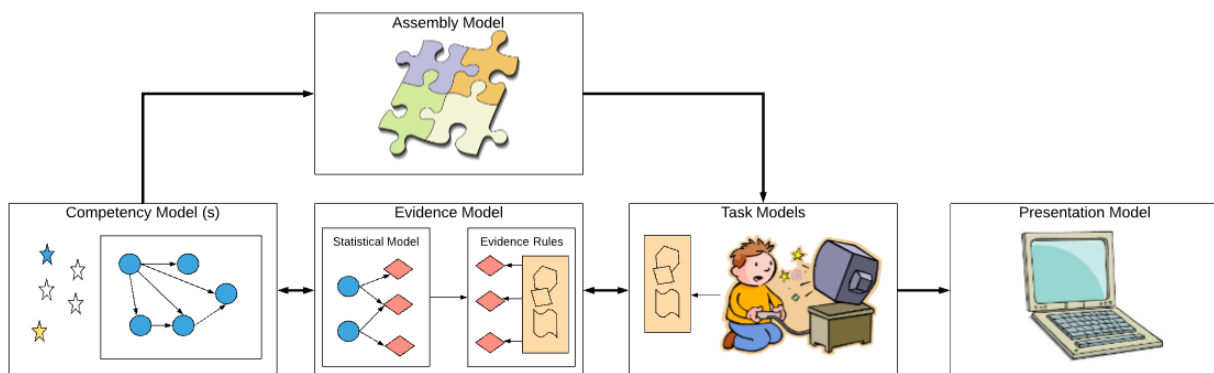
Stealth assessments involve the collection of learning evidence without learners' awareness (i.e., while they are playing a game, see Shute & Ventura, 2013). For young learners in particular, assessments embedded in games may address some of the challenges associated with assessing young learners. Bailey (2017) notes that assessing younger learners can be challenging due to their susceptibility to testing fatigue, anxiety, and wandering attention, among other factors. McKay (2006) further supports this arguing that assessments designed for younger learners must address motivational needs, provide contextual support, and adopt an iterative approach that draws from multiple sources of evidence. Stealth assessments address many of these challenges by embedding assessments in digital games which are highly contextualized and engaging while also allowing for data to be collected over time.

A central element of stealth assessment is the Evidence Centered Design (ECD) framework (Mislevy et al., 2003; Shute & Ventura, 2013). ECD formalizes the process for assessments as evidentiary arguments including claims to be made about students' competencies and the evidence supporting those claims. In the context of digital games, evidence of students' competencies can be derived from game data.

ECD is a multilayer approach comprising five layers: (a) domain analysis, (b) domain modeling, (c) the conceptual assessment framework, (d) assessment implementation, and (e) assessment delivery. In layers 1 and 2, the focus is on the purposes of the assessment, the nature of knowing, and the structures for observing and organizing knowledge. In the third layer, assessment designers focus on the competency model (for example, what skills are being assessed), the evidence model (for example, how skills are measured), and the task model (for example, how situations are designed to elicit behaviors/evidence). These aspects of assessment design are interrelated. The models promote a shift in assessment design towards a focus on the process of task design and its relationship to statistical interpretation, rather than simply analyzing and interpreting data out of context (Mislevy et al., 2003).

Figure 2

Conceptual Assessment Framework



Of relevance to the present study is the third layer, the conceptual assessment framework. This framework focuses on the assembly of the entire assessment and the processes needed to implement (stealth) assessments by generating a test blueprint that specifies the scoring system, statistical models, and delivery processes (Mislevy & Riconscente, 2005) for the assessment. [Figure 2](#) provides a visualization of the five components that make up the conceptual framework for stealth assessments and visualizes their interconnected nature.

The first model is the *competency model*, which has also been called the student model. This model identifies what the assessment is measuring (Mislevy & Riconscente, 2005). The second model is the *evidence model*, which provides the technical details for how the assessment measures the competency model. The evidence model contains two parts, the evidence rules and the statistical model. The evidence rules refer to how information is decoded, in this case from a computer, to match with the student variables. For instance, the process by which time on reading tasks was calculated and eventually discretized for analysis in the present study is one example of an evidence rule. The statistical model explains how evidence is associated with the competencies outlined in the competency model. For example, the statistical model might state that 80% or higher on a grammar quiz has been designated as mastery of the grammatical concept for the purposes of interpreting assessment results.

Third is the *task model*. This model focuses on the design of tasks that will elicit the competency variables above. Simply stated, it describes where and how we measure the targeted skills and/or knowledge. Tasks should be described by how they are presented, types of outcomes elicited, and in terms of variables associated with the task features (for example, difficulty). This model considers all possible behaviors that may be elicited from a task and if those behaviors can then be mapped onto the targeted competencies.

The *assembly model* describes how the competency, evidence, and task models work together to provide sufficient information for a viable assessment (Almond et al., 2014). The last model is the *presentation model*, which describes how the game is presented and played on certain devices (e.g., laptop, tablet) and how it is presented to the learners (e.g., as an assessment or as a game to play).

The Present Study

The primary focus of the present study was to develop a stealth assessment to evaluate young Chinese DLI learners' L2 reading comprehension. The five components of the stealth assessment within the ECD's conceptual assessment framework as they were operationalized in the present study are presented in [Table 1](#).

Table 1*Stealth Assessment Conceptual Assessment Framework Model Operationalizations*

Conceptual Assessment Framework Model	Conceptual Assessment Framework Questions	Operationalization for the present study
Competency	What are we measuring?	Reading comprehension
Evidence	How do we measure it?	Bayesian belief networks are used to describe the probability of reading comprehension given the evidence of student performance on reading tasks (for example time spent on reading, support used, number of correct responses, and so forth).
Task	In what situation(s) do we measure it?	Short in-game reading prompts supported by a glossing system
Assembly	How much information is needed?	Students played the game for approximately 200 minutes across 8 sessions and were exposed to 550 in-game texts on average.
Presentation	How is the game presented to the learner?	Students played the game on MacBook computers in their regular classroom setting.

The present study was guided by the following research question:

To what extent do students' performances on in-game tasks as measured by stealth assessments correlate significantly with external reading measures?

Method

This study is part of a larger project. In Poole et al., (2022) we describe the design of the game and the learning outcomes that were associated with the game. Poole and Clarke-Midura (2023) discuss the role of educational data mining in extracting data from the game and exploring potential relationships between those data. In this article, we illustrate how BBNs can be used to assess reading proficiency while learners play a language-related game.

Participants and Instructional Context

Data were collected in two sixth-grade Chinese DLI classrooms in the United States. The two classes had 19 and 21 students respectively. Students ranged in age from 10 and 12 years old ($M = 11.05$). Consent was not received from four students (1 in Class A, 3 in Class B). In addition, two students in Class B did not take the pre-test, and one student in Class A and B did not take the post-test. Only students who completed all components of the consent process and study were included in our analysis ($N = 34$). The students were using simplified Chinese characters while learning Mandarin Chinese.

Procedures

The game was played on a set of MacBook computers that were provided by the research team. All data was uploaded to a cloud-based database after each gameplay. The students played the game for four weeks, twice per week, with each session lasting approximately 25 minutes (approximately 200 total

minutes of gameplay per student). A detailed overview of individual and group-level descriptive statistics is available in Poole and Clarke-Midura (2023).

Data Collection

Data for this study came from four sources: pre- and post-vocabulary and reading comprehension tests, teacher evaluations of student reading comprehension, and log files that captured in-game actions.

Vocabulary assessment

The pre- and post-vocabulary tests (see [Appendix B](#)) consisted of 45 words from the game. On the assessments, learners were presented with each word in character form (mostly two-character words) and were first asked to indicate whether they (a) knew the word, (b) thought they knew the word, or (c) were just guessing. Next, learners were prompted to enter the pinyin and the English meaning. These tests were scored by awarding 1 point for correct pinyin and 1 point for correct English definitions. Further, half points were awarded for partially correct answers. For example, if a learner correctly identified the pinyin or English for one of the characters but not both, half a point was awarded. Awarding partial points for vocabulary knowledge was viewed as valuable given past research that has noted the non-linear trajectory and gradual accumulation of vocabulary knowledge (see Schmitt, 2010). Both pre- and post-vocabulary assessments had Cronbach's alpha levels of over 0.90. Results from both tests are presented in [Table 2](#).

Table 2

Vocabulary and Reading Comprehension Descriptive Statistics

	Pre-Test					Post-Test				
	<i>M</i>	<i>SD</i>	α	Skewness	Kurtosis	<i>M</i>	<i>SD</i>	α	Skewness	Kurtosis
Vocabulary (<i>N</i> = 32)	18.31	14.38	0.95	1.45	2.20	27.95	17.97	0.96	1.14	1.08
Reading Comprehension (<i>N</i> = 32)	3.41	2.56	0.74	0.39	-0.98	4.72	2.43	0.60	0.15	-0.78

Reading Comprehension Assessment

The reading comprehension test was adapted from the Chinese YCT ('Youth Chinese Test', see Chinese Test Services, n.d.) which is an official Chinese proficiency assessment administered by the PRC's Center for Language Education and Cooperation (中外语言交流合作中心, formerly Hanban 国家汉办) and is used regularly in U.S.-based Chinese DLI programs. The test item formats were therefore already familiar to the students. The reading comprehension test consisted of ten multiple-choice items (see [Appendix A](#)). Although the format of the reading comprehension test was based on the YCT, the content was adapted to resemble texts like those in the game. To reduce priming effects, items were randomized in both the pre- and post-assessments. Further, although the sentence structures across tests remained the same, the content, and thus the answers, changed from pre- to post-assessments. Cronbach's alpha for the pretest was 0.74 but was only 0.60 on the posttest, which is somewhat lower than the ideal level of reliability and is a limitation of the present study.

Informal Reading Comprehension Assessment

Students' current (6th grade) and most recent past (5th grade) teachers both holistically rated each student based on their perceptions of students' overall reading comprehension abilities. Teachers were instructed

to rate on a scale from 1 to 5, with 5 being most proficient and 1 being least proficient. The teachers' two scores were then correlated to check reliability and a significant correlation was found ($r = 0.82$, $t(34) = 8.52$, $p < .001$). These scores were then averaged to create a composite score which was used as a second piece of validity evidence for the stealth assessment (see Data Analysis and Results). One of the arguments for using stealth assessments relies on the idea that traditional multiple-choice items usually lack contextualization and that they occur at one point in time, and are therefore possibly not robust measures, especially of language ability. Teacher ratings are based on evidence accumulated from multiple data sources over long periods of time, and we therefore considered these ratings a valuable potential source of validity evidence.

Gameplay data

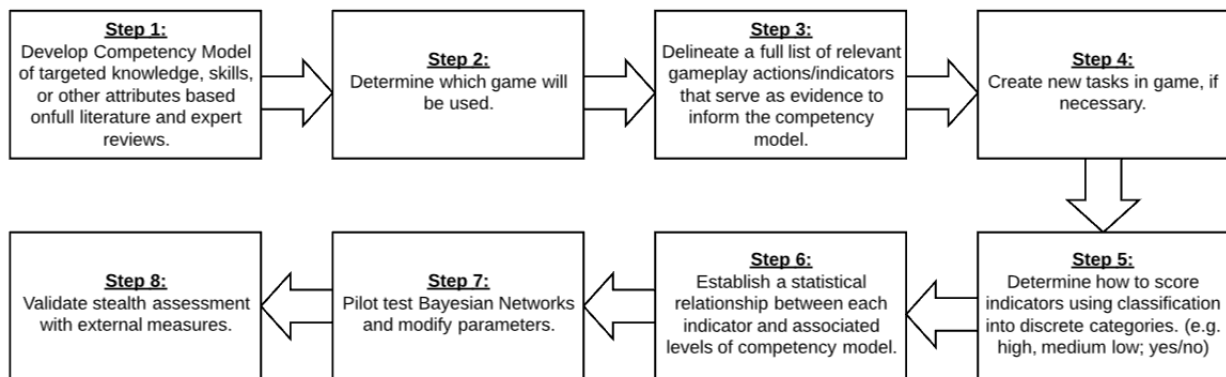
Finally, data from log files that captured players' in-game actions and reading behaviors were collected for all students. Specifically, players' movements and choices along with corresponding timestamps were saved to a JSON file after each session's gameplay and then automatically uploaded to a cloud-based database each time a player saved the game. After the study was completed, all JSON files were downloaded from the cloud-based database and then cleaned for analysis. More information about this process can be found in Poole and Clarke-Midura (2023).

Data Analysis

The stealth assessments were designed following an 8-step framework (see Figure 3), focusing on identifying key variables, selecting an appropriate game, developing an evidence model using in-game actions as indicators, and establishing a statistical relationship through Bayesian Networks (Shute et al., 2017).

Figure 3

Stealth Assessment Creation Framework (adapted from Shute et al., 2017)



Due to limited space, we focus on steps 3 and 5 of our analytical approach and refer interested readers to the Appendices for further information. Step 3 calls for the identification of the in-game actions that serve as indicators of the outcome of interest (in our case, reading comprehension). Table 3 provides a list of the identified indicators that were used, some of which were discretized (i.e., continuous variables which were collapsed into discrete variables, a necessary step for BBN analysis). Only variables that affected how the in-game texts were read were included in the present study (i.e., other gameplay variables that are not relevant to reading were not analyzed).

Step 5 involves the discretization of variables for analysis. The first variable, *time*, is a continuous variable that measures the number of seconds a player uses when reading an in-game text. Any reading time that was less than 2 seconds was identified as *fast*, or as skipping the text. Any reading time that was more than 2 standard deviations above the group's mean reading time (i.e., > 6 seconds) was marked as

slow. Anything in between (i.e., 2 to 5 seconds) was seen as optimal reading time given the average length of the texts and the typical reading speed of the participants.

The second variable, *respond*, is a categorical variable indicating whether the text that was being read prompted students to respond. This was seen as an important indicator since past research has shown that when students are asked to respond to a text, they are more likely to both pay (more) attention to the text and confirm their understanding of it (Poole et al., 2018).

The next variable, *Sent_K*, represents the length of a sentence. Initially, *Sent_K* was a continuous variable counting the total number of characters in a text. To make it more manageable, we applied K-means clustering, dividing texts into three categories: short, medium, and long. K-means clustering created a balanced distribution, defining short texts as 0-6 characters, medium texts as 7-30 characters, and long texts exceeding 30 characters. *LookUp* is a categorical variable that indicates whether the glossing system for a word was used. *Vocabulary* is also a categorical variable that indicates whether a given text contains a word that an individual student knew on the pretest. There are three levels of this variable. *Known* refers to a text that contains at least one target vocabulary word that the student previously knew. *Unknown* refers to a text that contains one or more vocabulary words that the student did not know at the time of the pretest, and *Both* indicates a text that contains both previously known and unknown words. *ReadComp* was used to track reading comprehension. Some of the texts in the game have clear right and wrong responses. For these texts, the variable *ReadComp* was used to track correct and incorrect responses. However, in the BBN, other texts which did not require a specific ('correct') response are predicted using expert-driven parameters (see [Table 3](#)).

Table 3

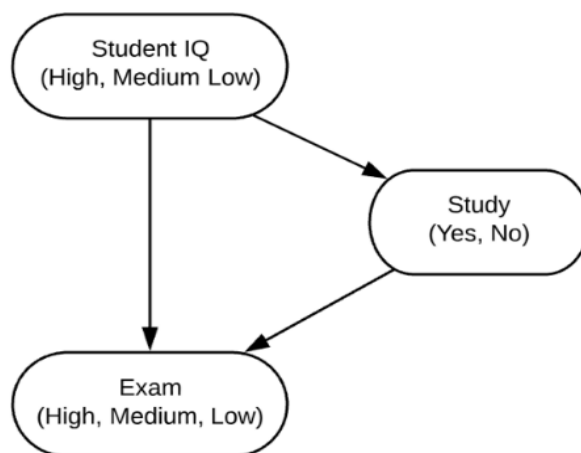
In-Game Variables Used for Analysis

Variable	Description	Collected Format	Discretized Categories
Time	Time spent reading text in the quests	Continuous	Fast Medium Slow
Respond	Whether or not the text being read required a response	Categorical	Respond NoRespond
Sent_K	Length of the sentence	Continuous	Short Medium Long
LookUp	Whether or not a word was looked up	Continuous	No Look Up Look Up
Vocabulary	Whether or not a text contains a known word	Categorical	Known Unknown Both
Duplicates	How many times the same text has been read	Continuous	Origin Less than three times More than three times
ReadComp	Outcome variable indicating if a text was read/understood	Categorical	Read/Understood Not Read/Understood

Finally, the *Duplicates* variable tracks how many times a student has read a text. This variable was originally continuous and was converted to categorical by first identifying the first time a text was read (*Origin*), then identifying if the text was read less than three times, and finally, if a text was read more than three times (See [Appendix C](#) for frequency data on these categories). For a more detailed description of the 8-step process used to create this stealth assessment, see [Appendix D](#).

Bayesian Belief Networks

Bayesian belief networks are statistical models that reflect how a set of variables are related by probabilities. [Figure 4](#) shows an example of a simple BBN.

Figure 4*Simple BBN Example*

In this example, there is a student with either a high, medium, or low IQ, a variable that determines whether the student studies or not, and an outcome variable (exam), for which either a high, medium, or low result is possible. When data are entered into the model, Bayesian networks determine the likelihood that a student with a high IQ will score low on an exam, or the probability that a student with a low IQ who chooses to study receives a medium score on the exam. Bayesian networks are based on Bayes' Rule, which states that the probability of an *outcome A*, given *evidence B*, is equal to the probability of *B*, given *A*, multiplied by the probability of *B*, divided by the probability of *A* (See Figure 5).

Figure 5*Bayes Rule*

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Note. $P(A|B)$ represents the probability of A given B occurred

Bayesian networks use interconnected webs of conditional probability tables (CPTs) represented by nodes within a simple causal graphical structure. *Influence* diagrams (see Figure 6), made up of nodes (the CPTs) and arrows, display the direction of influence each node has on other nodes. In the example above, the exam score is influenced by both studying and student IQ, and whether how a student studies is further influenced by their IQ. The CPTs provide probabilities regarding the belief that a particular state will be present given the evidence (Chen & Pollino, 2012). For instance, given the evidence that a student studies and has a high IQ, the CPT for exam score in the previous example might state that there is an 85% chance for a high score, a 10% chance for a medium score, and a 5% chance for a low score. Evidence is then entered into the model and probabilities are propagated throughout each node. Thus, if a student with a high IQ who studied received a medium score, the final CPT for Exam score will update to reflect this new information.

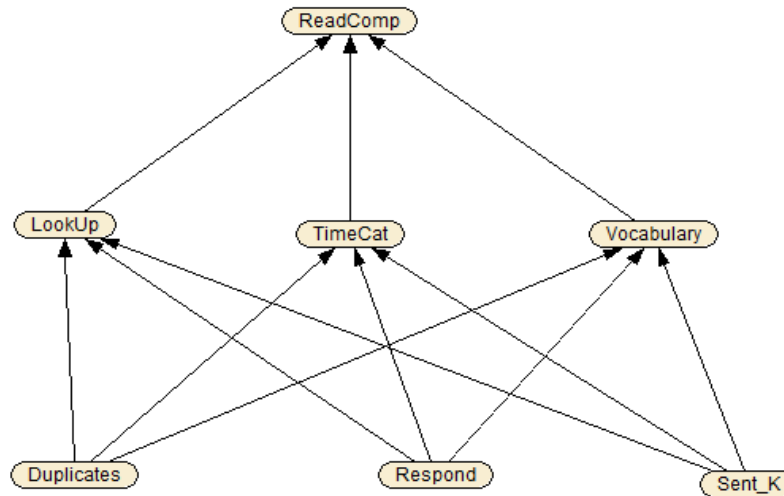
Recent BBN software has reduced the technical knowledge previously needed to build networks and made interpreting BBNs easier. In this study, the software *Netica* (www.norsys.com) was used to

construct the BBNs. Although *Netica* makes the technical side of creating BBNs easier, the conceptual development, calibration, and validation of BBNs still require rigorous detail and documentation. Marcot and colleagues' (2006) detailed guidelines for building and validating BBNs were used in this study. Speaking broadly, there are three steps involved: establishing the BBN structure, defining the parameters, and then testing/validating the model.

Marcot et al. (2006) broke the process down into three stages. The first step is to identify a structure. The BBN structure can either be defined by an expert via literature review or can be determined through data mining approaches via available data (typically very large datasets). In most cases, the expert-defined model is preferred as the data-induced model can lead to overfitting with results that make little sense (Marcot et al., 2006). Figure 6 is the expert-driven model we designed for the present study.

Figure 6

Expert Model BBN



In our expert-defined model, *LookUp*, *Time*, and *Vocabulary* were presumed to have a direct effect on reading comprehension. *Duplicates* is assumed to affect *LookUp*, as the more times a learner sees a similar text the less likely they are to look a word up. Similarly, *Duplicates* influences *Time*, as repeated exposure to texts should decrease the amount of time spent on a text. Lastly, the more times a learner is exposed to a text, the more *Vocabulary* they will be exposed to. *Respond* refers to a text that prompts a learner to respond to a question. Thus, *LookUp* should be more prominent, *Time* should be longer, and *Vocabulary* should be increased given the increased amount of text with extra prompts. Similarly, *Sent_K* (the length of the text) should impact reading *Time* and *Vocabulary* given that longer texts will take longer to read and have more vocabulary. It was also assumed that longer texts may result in more use of the glossing feature, *LookUp*.

The next step in the refinement of a BBN-based analysis involves adding CPTs to each of the nodes and discretizing included variables as necessary (that is, for all continuous variables to be included in the model). Marcot et al. (2006) provided a list of suggestions for this step (see Table 4). Many of the suggestions were provided to keep the parameterization process manageable.

Table 4*Discrete Category Model Construction Steps*

Guidelines for Model Construction
Number of Parent nodes should be kept to three or fewer.
Parentless nodes should be those that have observable cases.
Intermediate nodes summarize major themes.
To the extent possible all nodes should be observable and quantifiable, in cases where they cannot the nodes should be carefully documented and explained.
Fewest discrete states possible, but enough to get precision.
Number of layers should be kept to four or fewer.
Model should be fully documented with the rationale for each node.

Note. Adapted from Marcot et al. (2006)

In the present study, all continuous variables were discretized according to the methods outlined above, with each resulting node containing two or three states. Further, all nodes, except for three parentless nodes, have three parents.

The next step in the BBN analysis involves creating *a priori* CPTs for each of the nodes. These *a priori* CPTs are an initial set of probabilities that represent the theorized relationship between each of the nodes. It is important to note here that once data is added to the model, the CPTs will adjust according to the data entered. Marcot et al. (2006) provided a few strategies for creating expert-defined CPTs when past data is not available (as is the case in this study). One approach is to set the CPTs to uniform value, so the probabilities are set with each state having an equal chance of occurring. Alternatively, “peg the corners” is a strategy in which the extreme cases are first defined (usually as 0% or 100%) and then the intermediate cases are set by adjusting all the remaining cases (Marcot et al., 2006, p. 3067). In our study, we used the “peg the corners” strategy to define the *Reading Comprehension* node which has *LookUp*, *TimeCat*, and *Vocabulary* as parent nodes. This was important because *Reading Comprehension* has limited (data) cases to validate and thus the outcomes will rely largely on the expert-defined CPTs (so these must be carefully defined and justified). The other six nodes were set to uniform values without expert input. Although adding expert input to set CPTs is typically recommended, this was not seen as a major issue given the large number of cases from our dataset available for each of these remaining nodes.

Table 5 shows the CPT that was created for the *Reading Comprehension* node. To define this table, we first started with all the *LookUp* equals *No* states. Then we compared vocabulary knowledge within each *Time* state. Thus, not knowing a vocabulary word and not using the glossing system to look up the unknown word was set as the lowest probability, a 5% chance of comprehending the text. We did not make this impossible (that is, 0%) as there is a small chance that a word was learned via images and had been seen multiple times already, and thus there is still a chance that it was read. Then for this subgroup (*NoLookUp* and *Fast* reading time), we made knowing a vocabulary word the highest probability for reading comprehension at 20% and both knowing and not knowing a word as an intermediate value (10%). Next, we adjusted all these scores for the *NoLookUp* with *medium* and *slow* reading times, making the medium times the most likely to be examples of reading comprehension and the slow reading times to be the intermediate (between fast and medium) in terms of reading comprehension probability. Finally, we copied the probabilities and pasted them into the *LookUp* equals *Yes* states and uniformly increased the probabilities of reading comprehension to reflect that using the glossing system is a strong indicator that the student is attempting to read the text.

Table 5*Reading Comprehension Conditional Probability Table*

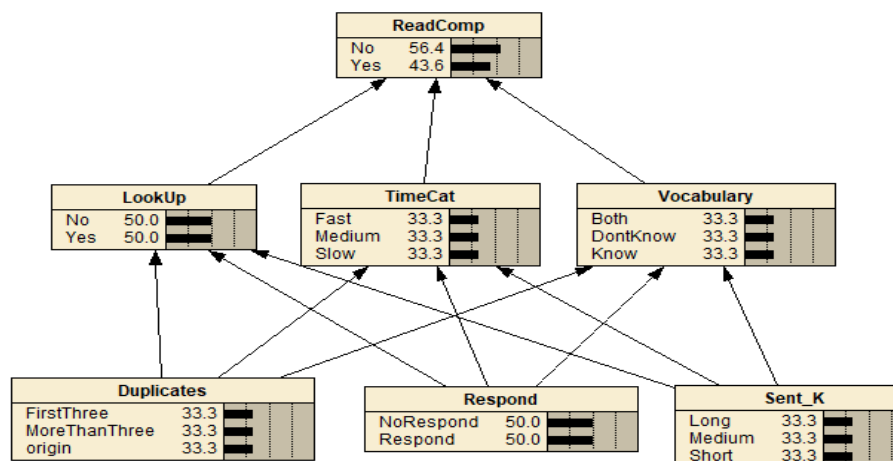
Reading Comprehension		LookUp	Time	Vocabulary
No	Yes			
90	10	No	Fast	Both
95	5	No	Fast	Don't Know
80	20	No	Fast	Know
45	55	No	Medium	Both
55	45	No	Medium	Don't Know
40	60	No	Medium	Know
70	30	No	Slow	Both
75	25	No	Slow	Don't Know
60	40	No	Slow	Know
65	35	Yes	Fast	Both
75	25	Yes	Fast	Don't Know
60	40	Yes	Fast	Know
25	75	Yes	Medium	Both
30	70	Yes	Medium	Don't Know
20	80	Yes	Medium	Know
40	60	Yes	Slow	Both
45	55	Yes	Slow	Don't Know
35	65	Yes	Slow	Know

Results

After adding all the CPTs and then compiling the BBN, the following network was produced ([Figure 7](#)). Note that all the parents of *Reading Comprehension* and the parentless nodes are set to the probabilities that give each state an equal probability of occurring. *Reading Comprehension* is set so that there is a 43.6% chance that a text is understood. This is the baseline prediction (the base model, constructed using expert input and CPTs as previously described) without adding any cases (that is, data) to update the CPTs.

Figure 7

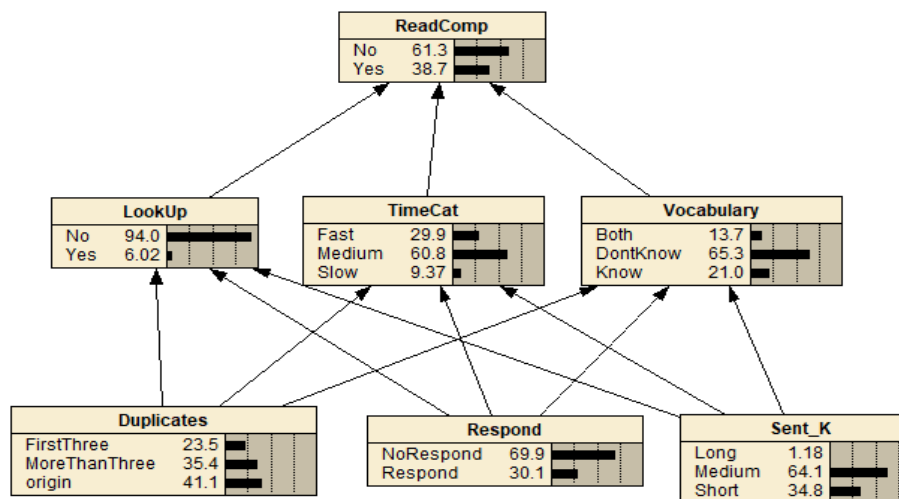
Initial Compiled BBN (Expert-Driven Model with CPTs Defined, Prior to Adding Data)



The next step to validate BBNs involves adding cases to automatically update the CPTs. Therefore, 80% of gameplay data from our dataset was added to the model. The remaining 20% of the data was reserved for a subsequent step in the validation process. The updated CPTs from this step are reflected in [Figure 8](#).

Figure 8

Overall BBN



With data added to the BBN, it is necessary to add evidence to the model to confirm that it is functioning properly. This step involves assuming that one of the states in a node has a 100% chance of occurring and then exploring how this impacts the other CPTs. In *Netica*, one can simply click on the node and set its probability level to 100%. For example, if *LookUp (Yes)* is clicked (see [Figure 9](#)), it is assumed that *Reading Comprehension* will increase along with *Origin*, as learners are more likely to look up a word the

first time they see a text. **Figure 9** confirms this finding by showing that when students did look up a vocabulary word, their chances of reading a text increased to 61.9% and the likelihood that they were reading an original text increased to 67.4% from 41.1%. In addition, fast texts or texts that are skipped are also reduced, suggesting that when students look up a word, they are indeed reading a text.

Figure 9

Adding Evidence: LookUp

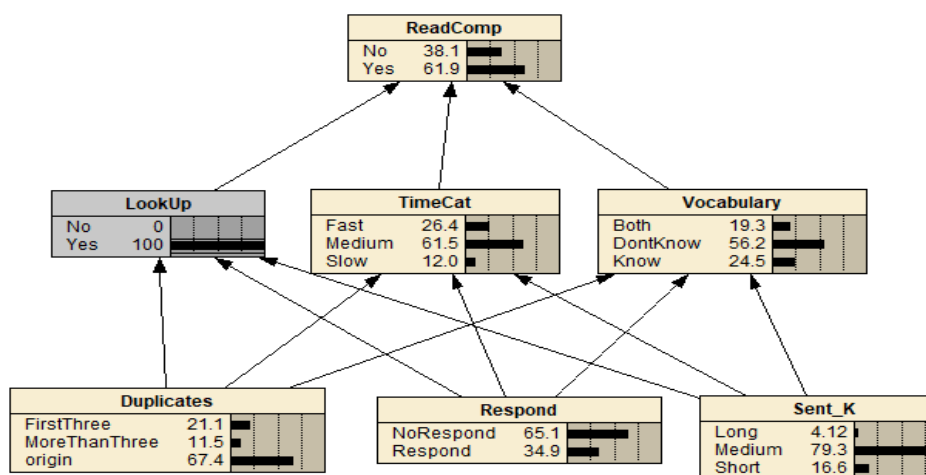
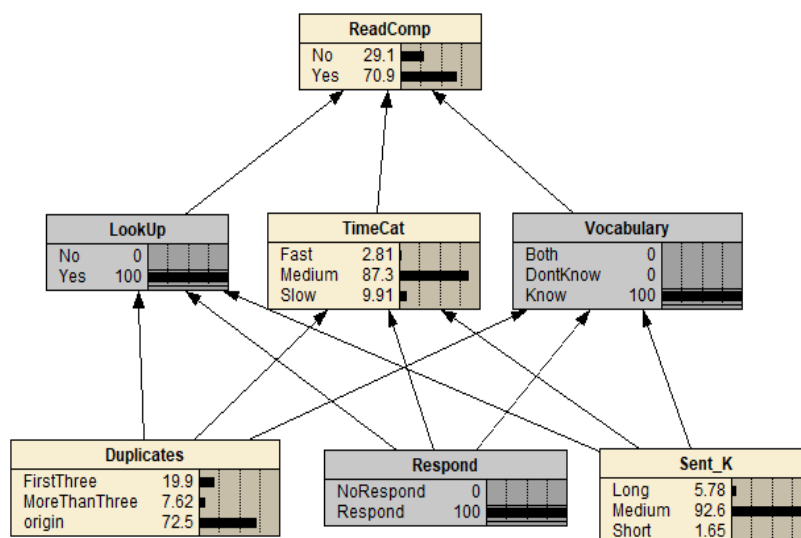


Figure 10 shows that when a student looks up a word, knows a vocabulary word in the text, and responds to a prompt, there is a 70% chance that the student will read the text. Further, there is a 72% chance that comprehension will occur the first time a student encounters the text and a 92% chance that it will happen for a medium-length sentence.

Figure 10

BBN with Additional Evidence



The next method we used to validate the BBN's performance involved the use of a confusion matrix (see Table 6). A confusion matrix is a table summary comparing the BBN-predicted versus actual observed results (from the dataset) to succinctly evaluate the model's proportion of correct and incorrect predictions. Diagonal cells (that is, the shaded cells in Table 6) show correct classifications. Comparing these allows the assessment of model performance. Put simply, the more reliable a model is, the more correct classifications will be observed in the confusion matrix. To construct this matrix, the previously unused 20% of the cases from gameplay data which were initially withheld when the BBN was updated with cases were put into the model. These data allow us to determine the accuracy of the model to predict reading comprehension. It's important to note that a total of 3% of the in-game texts ($N = 606$) contained prompts with definitive (in)correct answers. Data from this subset of texts for all students were used to test the accuracy of the model. Overall, the model had an error rate of 22.3%.

Table 6

Confusion Matrix

Predicted Reading Comprehension		Actual Reading Comprehension	
No	Yes	No	Yes
2	28	No	Yes
5	113	Yes	Yes

The final step to evaluate the efficacy of the BBN involved applying the model to each student's gameplay data individually (up to this point, the model used aggregate data). The CPTs from the class-wide BBN, therefore, were applied as the *a priori* CPTs for each student BBN, and then student case data were added to the model to automatically update the competency model's CPTs. Using this model, the predicted probability of reading comprehension for each student is obtained. These individual prediction values are then correlated with student scores on the external reading assessment and teacher assessments to provide (concurrent) validity evidence of the model's effectiveness at assessing reading comprehension.

The average probability scores for the students are shared in Table 7. Students were predicted to have read a text an average of 47% of the time, used the glossing tool on 11% of the texts, and used an ideal reading time about 57% of the time. It is important to remember that these are predicted averages based on performance within the game. These reading comprehension probability scores were then correlated with both their post-test scores on the external reading comprehension test and with the teachers' composite assessments of students' overall reading levels. The BBN reading comprehension scores for the students were significantly correlated with the external reading comprehension test ($r = 0.52$, $t(32) = 3.45$, $p = .001$). Further, there was a stronger correlation with the teacher scores ($r = 0.66$, $t(32) = 4.99$, $p < .001$).

Table 7*Probability Score Averages*

	Mean	SD	Min.	Max.
Reading Comprehension (Yes)	0.47	0.04	0.40	0.57
LookUp (Glossing Tool Use)	0.11	0.06	0.03	0.27
Respond (Prompted to Respond)	0.31	0.03	0.24	0.36
Origin (First time Seeing a Text)	0.43	0.09	0.24	0.64
First Three (Seeing a Text)	0.24	0.03	0.18	0.30
More than three (Seeing a Text)	0.33	0.10	0.11	0.54
Sentence Length (Short)	0.33	0.06	0.23	0.47
Sentence Length (Medium)	0.64	0.05	0.52	0.75
Sentence Length (Long)	0.02	0.02	0.00	0.06
Time (Fast)	0.31	0.04	0.23	0.40
Time (Medium)	0.57	0.04	0.46	0.67
Time (Slow)	0.12	0.03	0.07	0.19
Vocabulary (Both)	0.19	0.07	0.06	0.33
Vocabulary (Know)	0.25	0.12	0.10	0.61
Vocabulary (Don't Know)	0.55	0.14	0.26	0.83

Discussion

The primary objective of this study was to examine the extent to which stealth assessments using game-based data could accurately predict individual L2 Chinese students' overall reading abilities. The BBNs leveraged students' use of the in-game glossing tool, average time spent reading texts, and prior vocabulary knowledge to generate reading comprehension probability scores which were found to be significantly correlated with multiple external assessments. This finding aligns with previous research in other content areas which also found significant correlations between stealth assessment results and external measures (e.g., Shute et al., 2016). Further, the BBN we constructed had an error rate of 22.3%. While there is no hard line on how much error is acceptable (Marcot et al. 2006), much can be learned about how the stealth assessment works by examining the confusion matrix. In the case of this study, of the 148 examined texts with a definite outcome, 141 were identified as having been read and understood, while only 7 were identified as not understood. Only 23 texts were misclassified, due to readers providing an incorrect answer. These findings suggest that our stealth assessment might be predicting when students *attempted* to read (i.e., reading engagement) rather than precisely whether they read and understood a text. Reading engagement has previously been found to be significantly correlated with reading comprehension (Wigfield et al., 2008).

Interestingly, the BBN was more strongly correlated with teachers' informal assessments than paper-based measures of reading comprehension ($r = 0.66$ and 0.52 , respectively). While the teacher scores were based on holistic judgment rather than specific recurring documentation using any assessment instrument, they were likely longitudinally derived judgments based on multiple observations of learners' reading. This could be interpreted as evidence that the BBN is acting closer to teachers' regular practice of observing student performance and gradually forming an idea of their ability levels, which in many ways is similar to a stealth assessment in that students are not always aware of their teachers' evaluation of their performance taking place when engaged in classroom learning.

Our results suggest that BBNs and stealth assessments embedded in L2 games are potentially promising tools to monitor and document L2 learners' proficiency in L2 (Chinese) reading. Extracting data from L2 games and analyzing it using BBNs may offer insight into learners' overall reading ability by synthesizing a greater quantity and quality of evidence than would be possible in other settings. This data is a meaningful finding in the context of assessment, in which one might argue that one-on-one teacher-to-student individual assessment is the ideal but is highly inefficient and therefore underutilized in education. Therefore, given our finding that game-based stealth assessments may provide similar results to teachers' longitudinal perceptions of student abilities, stealth assessments may be both an optimal and reliable source of assessment data for language teachers, especially with younger learners (i.e., grade school contexts) with large class sizes.

Limitations and Future Research Directions

This study has a few important limitations. First, the sample size was relatively small for this type of study. Ideally, the parameters that were used for the student models would have come from a larger population-level dataset instead of the class from which data were gathered. This is important because the model assesses the learners based on their relative performance compared to their classmates. Generalizability, therefore, is limited. Second, the study focused on a limited subset of words within the game. Students' pre-test knowledge of this subset of words was used to determine whether they were reading texts containing unfamiliar words. More robust vocabulary knowledge would be a more ideal operationalization of (un)known words and would likely improve the BBN's ability to predict reading comprehension. BBNs are capable of making inferences despite this lack of data, but more data would undoubtedly increase the accuracy of the models. Future studies could explore reading comprehension when students are engaging in a wider variety of tasks, which will provide more information to the BBN and further improve prediction accuracy. Additionally, the reliability of the external post-assessment was lower than ideal, and this was a relatively minimalist operationalization of reading comprehension, and thus findings should be interpreted conservatively.

Despite its limitations, the results of this study have several implications for future research. First, if in-game behaviors are indeed able to accurately predict students' L2 (reading) abilities (as our results suggest), game-based stealth assessments may be sites for robust research on multiple aspects of language development, including but not limited to reading, listening, and multimodal comprehension, but also productive abilities. To the extent that games are designed with tasks and prompts that elicit learner language, especially tasks that have a limited number of correct answers, stealth assessment data can be collected and analyzed similarly to our study's procedure. These results could then be compared to external measures of learners' language ability, including standardized measures as well as teacher judgments.

Furthermore, collecting data while learners are unaware of being assessed may provide a lower anxiety (and possibly more ecologically valid and representative of contextualized communication ability) measure of language skills. This is particularly important for language assessments which have been shown to elicit anxiety which can impact the outcome of the assessment (e.g., Salehi & Marefat, 2014). Future studies could collect and compare student language ability data on multiple assessments, including stealth assessments, as well as their levels of anxiety (or other similar affective variables) to further corroborate this possibility.

Conclusion

The goal of this study was to build and evaluate stealth assessments to assess L2 Chinese reading comprehension while young learners played a digital game in the classroom. Assessing L2 reading comprehension is difficult because the process is internal to the learner and reading depends on many factors, including vocabulary knowledge, grammar knowledge, and L1 reading abilities, among others (Jeon & Yamashita, 2014). Game-based assessments may help address some of the difficulties associated

with assessing L2 reading comprehension in young learners (McKay, 2006). This study serves as a first step towards exploring innovative approaches to L2 reading comprehension assessment, highlighting the need for assessment practices that are both data-driven and unobtrusive while learners engage with language in meaningful and enjoyable language learning tasks.

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Appendix A. Sample Reading Comprehension Test

蝙蝠喜欢吃水果。 ★蝙蝠会吃哪个？ A - 猪肉 B - 苹果 C - 米饭	兵马俑很古老，所以人们需要保护他们。 ★为什么兵马俑很古老？ A - 兵马俑太胖了 B - 因为生病了 C - 是中国古代的人
这个指南针坏了,所以指向西边。 ★指南针应该指哪边？ A - 东边 B - 北边 C - 南边	蜜蜂不能和大的动物战斗。 ★蜜蜂会和哪个动物战斗？ A - 老鼠 B - 熊 C - 狼

Appendix B. Sample Vocabulary Assessment Items

Chinese Character	I know the answer	I think I know the answer	I'm just guessing	Pinyin	English
蚂蚁					
蝙蝠					
欢迎					
龙					
鹰					
摘					
熊					
布					
找					
木头					
野猪					

Appendix C. Frequency of Discretized Variables

Variable	Description	Discrete Categories	Frequency
Time	Time spent reading text in the quests.	Fast	12,225
		Medium	6,739
		Slow	804
Respond	Whether or not the text being read required a response	Respond	6,182
		NoRespond	13,599
Sent_K	Length of the sentence.	Short	6,513
		Medium	12,875
		Long	393
LookUp	Whether or not a word was looked up.	No Look Up	18,439
		Look Up	1,342
Vocabulary	Whether or not a text contains a known word.	Known	1,908
		Unknown	5,904
		Both	1,382
Duplicates	How many times has the same text has been read.	Origin	8,153
		Less than three times	4,762
		More than three times	6,866
ReadComp	Outcome variable indicating if a text was read/understood.	Read/Understood	413
		Not Read/Understood	193

Note. Frequencies are totals for all students who played the game.

Appendix D. Detailed Description of Stealth Assessment Development

The stealth assessments in this study were designed using the eight steps outlined in [Figure 3](#), which were adapted from Shute et al. (2017).

Step 1: Identify Key Variables. Step 1 calls for the identification of a competency model based on literature review and expert feedback. The present study identified Chinese reading comprehension as the skill to be assessed. Past research exploring factors that affect L2 Chinese reading comprehension emphasized reading fluency, vocabulary knowledge, as well as character recognition as strong determinants for reading comprehension (Shen & Jiang, 2013). Further, a previous study with a similar participant sample noted the value of metacognition, and specifically self-regulation skills, while reading (Poole et al., 2018). Although word segmentation was also identified as an important skill, the tasks and game design used in this study did not allow this to be included as a variable for analysis.

Step 2 and Step 4: Building the Task Model. These steps are concerned with selecting a game and creating tasks that elicit evidence for the desired skills. The game being used is *Legend of the Dragon* (see Poole et al., 2022 for a detailed description). This game involved an open-world environment through which learners could explore and interact with non-player characters (NPCs) with autonomy. Players completed tasks on their journey to save the last dragon of China. Further, the game provided multiple L2 supports, such as a glossing system, “battle cards” with images and pinyin support, and a quest bar to remind and direct learners toward goals. The primary variable being assessed by the stealth assessment was learner engagement with and comprehension of texts within the game.

Step 3: Developing Evidence Model. This step calls for the identification of the in-game actions that serve as indicators of the outcome of interest (reading comprehension). [Table 3](#) provides a list of the identified indicators that were used. Only variables that affected how the in-game texts were read were included in the present study.

Step 5: Transform variables. This step calls for the discretization of the indicators from the previous step. [Table 3](#) displays the key indicators for reading comprehension and how they were discretized.

The first variable, *Time*, is a continuous variable that measures the number of seconds that a player uses when reading a text. At first, this variable was divided into fast, medium, and slow reading times by identifying the group-level mean, and then one standard deviation above and below the mean as slow and fast reading times. However, this did not make sense for a few reasons. First, there were some outliers in which some students used up to 40 seconds or more to read a text. These times inflated the mean but represented construct-irrelevant variance (i.e., were not due to slow reading speed, but usually due to students becoming distracted, for example). To fix this issue, first, outliers were temporarily removed and then a new mean and standard deviation were calculated. As a result, any reading time that was less than 2 seconds was identified as fast, or as skipping the text. Then any reading time that was more than 2 standard deviations above the mean (> 6 seconds) was marked as *slow*. These were typically moments when a learner was seeking external help or had gotten distracted. Anything in between, so 2 to 5 seconds was seen as optimal reading time given the average length of the texts.

The second variable, *Respond*, is a categorical variable referring to whether or not the text that was being read prompted students to respond. This was seen as an important indicator for two reasons. First, past research has shown that when students are asked to respond to a text, they are more likely to (a) pay (more) attention to the text, and (b) confirm their understanding of the text (e.g., Poole et al., 2018). Given that students may re-read or focus more attention on a text, it makes sense that their reading times will also be impacted, and thus this variable will likely affect reading times. The two categories are *Respond* and *No Respond*.

The next variable, *Sent_K*, represents the length of a sentence. Initially, *Sent_K* was a continuous variable counting the total number of characters in a text. To make it more manageable, we applied K-means clustering, dividing it into three categories: short, medium, and long texts. Originally, we used a similar

process as with time to categorize sentence length, but this led to an imbalance, particularly with numerous short texts containing only one word—commonly found when players picked up items. Switching to K-means clustering created a more balanced distribution, defining short texts as 0-6 characters, medium texts as 7-30 characters, and long texts exceeding 30 characters.

The next variable, *LookUp*, is a categorical variable that indicates whether or not the glossing system for a word was used. *Vocabulary* is also a categorical variable that indicates whether or not a given text contains a word that an individual student knew on the pre-test. There are three levels of this variable: *Known* refers to a text that contains at least one target vocabulary word that the student previously knew. *Unknown* refers to a text that contains one or more vocabulary words that the student did not know at the time of the pretest, and *Both* indicates a text that contains both previously known and unknown words. This classification of texts allows us to identify opportunities when students could have used the in-game glossing tool to support reading comprehension (i.e., understanding an unfamiliar word in a text they encountered) and whether or not they actually used the glossing tool.

The next variable *ReadComp* was used to track reading comprehension. Some of the texts (see [Table 5](#)) in the game have clear right and wrong responses, for these texts the variable *ReadComp* is used to track right and wrong responses. However, in the BBN, other texts which did not require a specific ('correct') response are predicted using the expert-driven parameters within the BBN (see below).

Finally, the *Duplicates* variable tracks how many times a student has read a text. This variable was originally continuous and was converted to categorical by first identifying the first time a text was read (*origin*), then identifying if the text was read less than three times, and finally, if a text was read more than three times (See [Appendix C](#) for frequency data on these categories).

Steps 6, 7, 8: Establish a Statistical Relationship. The last three steps are concerned with building, piloting, and validating a statistical connection between the evidence elicited and the competency model. Shute and colleagues (2017) recommend using Bayesian Networks for the statistical model in stealth assessments. Thus, in the following section, a brief introduction to this analysis is provided, followed by the procedures used to build, calibrate, and validate the analysis.

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